

SESSION 1 • LECTURE 1

Introduction to Thermodynamics: Systems, Properties & Processes

Detailed lecture notes — basic concepts, definitions, classifications and solved numericals

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Learning objectives. After this session you should be able to: (i) define a thermodynamic system, boundary and surroundings; (ii) classify systems as closed, open or isolated; (iii) distinguish intensive and extensive properties; (iv) define state, process, path and cycle; (v) distinguish point and path functions; (vi) convert between absolute, gauge and vacuum pressures.

1. What is Thermodynamics?

Thermodynamics is the science of **energy** and its transformations. It deals with heat and work interactions between a system and its surroundings, and with the properties of substances that store energy. The name comes from the Greek *therme* (heat) and *dynamis* (power) — historically, the effort to convert heat into motive power.

Classical (macroscopic) thermodynamics, which this course follows, describes matter using a small number of measurable bulk properties (pressure, volume, temperature) without tracking individual molecules. Statistical (microscopic) thermodynamics derives the same results from molecular behaviour.

- **Power generation:** steam and gas turbine plants convert the chemical energy of fuel into electricity; plant efficiency is fixed by thermodynamic limits.
- **IC engines:** petrol/diesel cycles, mean effective pressure and fuel economy are all first- and second-law calculations.
- **Refrigeration and air-conditioning:** pumping heat from a cold space to warm surroundings, within the limits set by the second law (Unit II).
- **Flow machinery:** turbines, compressors, nozzles and boilers are analysed with the steady flow energy equation derived in this unit.

2. System, Boundary and Surroundings

Every analysis begins by choosing the **system**: the quantity of matter or the region of space selected for study. Everything external to the system is the **surroundings**. The **boundary** is the real or imaginary surface, of zero thickness, that separates the two. The boundary may be *fixed* (gas in a rigid tank) or *movable* (gas beneath a piston); energy and, in some cases, mass are exchanged across it.

KEY POINT Always draw the system and mark its boundary before writing any equation — the boundary decides which heat, work and mass transfers enter the analysis.

Types of systems

Type	Mass transfer	Energy transfer	Other names	Examples
Closed system	No	Yes (heat and/or work)	Control mass	Gas in piston-cylinder; sealed pressure cooker
Open system	Yes	Yes	Control volume	Turbine, compressor, boiler, nozzle, pump
Isolated system	No	No	—	Ideal thermos flask; universe (system + surroundings)

An open system is analysed as a **control volume** (CV) — a fixed region in space — enclosed by a **control surface** through which mass carries its energy in and out. Most practical machines are open systems.

3. Properties of a System

A **property** is any measurable characteristic of a system whose value depends only on the state, not on how the state was reached: pressure p , temperature T , volume V , mass m , density ρ , internal energy U , and so on.

Basis	Intensive property	Extensive property
Dependence on mass	Independent of mass	Proportional to mass
If the system is divided in half	Value unchanged	Value halved
Examples	p, T, ρ, v, u, h	V, U, H, S, m
Notation convention	lower case (except p, T)	CAPITAL letters

Any extensive property divided by mass becomes a **specific** (intensive) property: specific volume $v = V/m$ (m^3/kg), specific internal energy $u = U/m$ (kJ/kg), specific enthalpy $h = H/m$ (kJ/kg). The quick test: mentally cut the system in half — if the value changes, the property is extensive.

4. State, Process, Path and Cycle

- **State:** the complete condition of a system, identified by the values of its properties. On a property diagram (p-V, T-s) a state is a single point.
- **Process:** the change of a system from one equilibrium state to another. Processes are named after the property held constant: *isobaric* ($p = C$), *isochoric* ($V = C$), *isothermal* ($T = C$), *adiabatic* ($Q = 0$).
- **Path:** the succession of states passed through during the process — the curve joining the end points on the diagram.
- **Cycle:** a sequence of processes that begins and ends at the same state; the net change in every property over a cycle is zero.

Point functions vs path functions

Feature	Point function	Path function
Depends on	End states only	The particular path followed
Differential	Exact: dp , dV , dT , dU	Inexact: δQ , δW
Cyclic integral	$\oint d(\text{property}) = 0$	$\oint \delta Q$, $\oint \delta W \neq 0$ in general
Examples	All properties (p , V , T , U , H , S)	Heat Q and work W

KEY POINT Heat and work are not properties. A system *possesses* energy; it never "contains" heat or work — these exist only while crossing the boundary.

5. Pressure: Absolute, Gauge and Vacuum

Pressure is the normal force exerted per unit area: $p = F/A$. SI unit: pascal ($1 \text{ Pa} = 1 \text{ N}/\text{m}^2$); working units: $1 \text{ kPa} = 10^3 \text{ Pa}$, $1 \text{ bar} = 100 \text{ kPa} = 10^5 \text{ Pa}$, $1 \text{ atm} = 101.325 \text{ kPa}$. Instruments read pressure *relative to the local atmosphere*:

$$p_{\text{abs}} = p_{\text{atm}} + p_{\text{gauge}} \quad p_{\text{abs}} = p_{\text{atm}} - p_{\text{vacuum}}$$

All thermodynamic relations require ABSOLUTE pressure

WORKED EXAMPLE 1.1 — Absolute, gauge and vacuum pressure

A pressure gauge on a compressed-air tank reads 120 kPa and a vacuum gauge on a condenser reads 66 kPa. The barometer reads 101.3 kPa. Find the absolute pressure in each vessel.

Solution.

1. Tank is above atmospheric: $p_{\text{abs}} = p_{\text{atm}} + p_{\text{gauge}} = 101.3 + 120 = \mathbf{221.3 \text{ kPa}}$.
2. Condenser is below atmospheric: $p_{\text{abs}} = p_{\text{atm}} - p_{\text{vac}} = 101.3 - 66 = \mathbf{35.3 \text{ kPa}}$.
3. Check: a vacuum reading can never exceed the barometric reading; $66 < 101.3 \checkmark$.

ANSWER Tank: 221.3 kPa | Condenser: 35.3 kPa

WORKED EXAMPLE 1.2 — Manometer reading

A U-tube mercury manometer ($\rho = 13,600 \text{ kg}/\text{m}^3$) connected to a gas pipe shows a height difference of 250 mm, with the gas side higher in pressure than the atmosphere. If $p_{\text{atm}} = 101.3 \text{ kPa}$ and $g = 9.81 \text{ m}/\text{s}^2$, find the absolute gas pressure.

Solution.

1. Gauge pressure from the column: $p_{\text{gauge}} = \rho gh = 13600 \times 9.81 \times 0.250 = 33,354 \text{ Pa} = 33.35 \text{ kPa}$.
2. $p_{\text{abs}} = p_{\text{atm}} + p_{\text{gauge}} = 101.3 + 33.35 = 134.65 \text{ kPa}$.

ANSWER $p_{\text{abs}} \approx 134.7 \text{ kPa}$

Note: For a column of liquid, $p = \rho gh$ with h in metres gives pascals directly.

WORKED EXAMPLE 1.3 — Specific volume and density

A tank of volume 0.75 m^3 contains 2.5 kg of a gas. Determine the specific volume and the density of the gas.

Solution.

1. Specific volume: $v = V/m = 0.75 / 2.5 = 0.30 \text{ m}^3/\text{kg}$.
2. Density: $\rho = 1/v = m/V = 2.5 / 0.75 = 3.333 \text{ kg/m}^3$.
3. Note v and ρ are intensive: any sample of this uniform gas has the same values.

ANSWER $v = 0.30 \text{ m}^3/\text{kg}$ | $\rho \approx 3.33 \text{ kg/m}^3$

6. Practice Problems (answers given)

#	Problem	Answer
P1	Classify: (a) a car battery discharging, (b) an air compressor in operation, (c) hot coffee in a perfectly insulated sealed flask.	(a) closed, (b) open, (c) isolated
P2	A gauge reads 1.75 bar at a place where the barometer stands at 98 kPa . Find the absolute pressure in kPa .	273 kPa
P3	A condenser operates at an absolute pressure of 8 kPa when $p_{\text{atm}} = 100 \text{ kPa}$. What will a vacuum gauge read?	92 kPa (vacuum)
P4	State whether each is intensive or extensive: temperature, enthalpy H , specific enthalpy h , volume, density.	T : int.; H : ext.; h : int.; V : ext.; ρ : int.

7. Summary

- Thermodynamics studies energy, and heat-work interactions between a system and its surroundings.
- Closed system: energy crosses, mass does not. Open system (control volume): both cross. Isolated: neither.
- Intensive properties are mass-independent; extensive properties scale with mass; specific properties are intensive.
- Properties are point functions (exact differentials); heat and work are path functions (inexact differentials).
- Use absolute pressure and absolute temperature in every thermodynamic relation.

Text: P.K. Nag, "Engineering Thermodynamics", 6th Edn., Tata McGraw-Hill — Ch. 1 (Introduction). Reference: Cengel & Boles, "Thermodynamics: An Engineering Approach", 8th Edn.